M1. From Parallel Arrays to a Class - Library Inventory

Week 3 OOP | Category B | Java solution with sample output

Objective

Implement the task using the classes, constructors, access control, references, or static members specified in the uploaded problem sheet.

Java Code

```
class BookInventory {
    String title;
    String author;
    int copiesAvailable;

    BookInventory(String title, String author, int copiesAvailable) {
        this.title = title;
        this.author = author;
        this.copiesAvailable = copiesAvailable;
    }

    void printEntry() {
        System.out.println(title + " by " + author + " - "
                + copiesAvailable + " copies available");
    }
}

public class Main {
    public static void main(String[] args) {
        BookInventory[] books = {
            new BookInventory("Clean Code", "Robert C. Martin", 3),
            new BookInventory("Effective Java", "Joshua Bloch", 5),
            new BookInventory("Refactoring", "Martin Fowler", 0),
            new BookInventory("Design Patterns", "GoF", 2)
        };

        for (BookInventory book : books) {
            book.printEntry();
        }
    }
}
```

Sample Output

```
Clean Code by Robert C. Martin - 3 copies available
Effective Java by Joshua Bloch - 5 copies available
Refactoring by Martin Fowler - 0 copies available
Design Patterns by GoF - 2 copies available
```

Concepts Covered

Classes and objects, constructors, arrays of objects, instance methods